C2.2 Bonding

Summary

A simple electron energy level model can be used to explain the basic chemical properties of elements. When chemical reactions occur, they can be explained in terms of losing, gaining or sharing of electrons. The ability of an atom to lose, gain or share electrons depends on its atomic structure. Atoms that lose electrons will bond with atoms that gain electrons. Electrons will be transferred between the atoms to form a positive ion and a negative ion. These ions attract one another in what is known as an ionic bond. Atoms that share electrons can bond with other atoms that share electrons to form a molecule. Atoms in these molecules are held together by covalent bonds.

Underlying knowledge and understanding

Learners should be familiar with the simple (Dalton) atomic model. They should be familiar with the principles underlying the Mendeleev Periodic Table and the modern Periodic Table including periods and groups, and metals and non-metals. Learners should have some knowledge of the properties of metals and non-metals including the chemical properties of metal and non-metal oxides wit respect to acidity.

Common misconceptions

Learners do not always appreciate that the nucleus of an atom does not change when an electron is lost, gained or shared. They also find it difficult to predict the numbers of atoms that must bond in order to achieve a stable outer level of electrons. Learners think that chemical bonds are physical things made of matte They also think that pairs of ions such as Na^+ and Cl^- are molecules. They do not have an awareness of the 3D nature of bonding and therefore the shape of molecules.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM2.2i ☑	estimate size and scale of atoms and nanoparticles	M1c
CM2.2ii	represent three-dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon	M5b
CM2.2iii	translate information between diagrammatic and numerical forms	M4a

Topic content			Opportunities to cover:		
Learning outcomes		**To include**	**Maths**	**Working scientifically**	**Practical suggestions**
C2.2a	describe metals and non-metals and explain the differences between them on the basis of their characteristic physical and chemical properties	physical properties, formation of ions and common reactions, e.g. with oxygen to form oxides		WS1.3f, WS1.4a	
C2.2b	explain how the atomic structure of metals and non-metals relates to their position in the Periodic Table				
C2.2c	explain how the position of an element in the Periodic Table is related to the arrangement of electrons in its atoms and hence to its atomic number	group number and period number	M1c	WS1.4a	
C2.2d	describe and compare the nature and arrangement of chemical bonds in: i. ionic compounds ii. simple molecules iii. giant covalent structures iv. polymers v. metals		M5b, M4a	WS1.4a	Make ball and stick models of molecules.
C2.2e	explain chemical bonding in terms of electrostatic forces and the transfer or sharing of electrons			WS1.4a	
C2.2f	construct dot and cross diagrams for simple covalent and binary ionic substances		M4a	WS1.4a	

C2.2g	describe the limitations of particular representations and models	dot and cross diagrams, ball and stick models and two- and three-dimensional representations	M5b	WS1.1c	
C2.2h	explain how the reactions of elements are related to the arrangement of electrons in their atoms and hence to their atomic number			WS1.1b, WS1.3f, WS1.4a	
C2.2i	explain in terms of atomic number how Mendeleev's arrangement was refined into the modern Periodic Table			WS1.1a, WS1.4a	